Code No: RR220305 **Set No. 1**

II B.Tech II Semester Supplimentary Examinations, Apr/May 2007
INSTRUMENTATION AND CONTROL SYSTEMS
(Common to Mechanical Engineering, Production Engineering and Automobile Engineering)

Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

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1. (a) Classify different transducers based on operation with examples.
   (b) Compare the following :
      i. Primary transducers Versus Secondary transducers
      ii. Active transducers Versus Passive transducers [8+8]

2. (a) Describe the different methods used for measurement of speed; describe their advantages and disadvantages.
   (b) Explain the construction, working and theory of a drag cup type tachogenerator. [16]

3. (a) Explain thermocouple protection materials and for what range they are used.
   (b) Explain the construction and working of
      i. Constant intensity optical pyrometer.
      ii. Variable intensity optical pyrometer.
   [12+4]

4. (a) Describe different sources of errors in U-tube manometer and how corrections can be applied to minimize these errors.
   (b) Explain how sensitivity can be increased by using inclined tube manometer. Describe its construction, advantages and limitations. [8+8]

5. (a) List out the importance of calibration of flow measuring instruments.
   (b) Explain the principle of operation off 'Laser Doppler' flow meter. [16]

6. (a) Explain the general theory of seismic instruments used for vibration/accelerator measurement.
   (b) Explain the working of a bonded strain gauge accelerometer with neat sketch. [8+8]

7. (a) Describe the properties of materials used for strain gauges.
   (b) Draw a simple strain gauge element and mark active and inactive direction
   (c) Explain the two arm and four arm conditions used for strain measurement. [4+4+8]

8. (a) The presence of disturbances is the main reason for using feedback control. Justify this statement. Discuss the commonly occurring disturbances in various control systems.
   (b) What are feedforward-feedback control system? Give some examples.
   (c) Explain with the help of a block diagram any one feedforward-feedback control system. [16]

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Code No: RR220305 **Set No. 2**

II B.Tech II Semester Supplimentary Examinations, Apr/May 2007
INSTRUMENTATION AND CONTROL SYSTEMS
(Common to Mechanical Engineering, Production Engineering and Automobile Engineering)

Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

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1. (a) Discuss briefly on the reliability of measurement systems. Explain how reliability improvement can be achieved.
   (b) "A pricise instrument need not be accurate" explain. [8+8]
2. (a) List the different types of speed-measuring devices. Explain with neat sketches, the construction and working of any two of them.
   (b) Write short note on the Tachogenerators. [8+8]
3. (a) List all industrial thermocouples giving their elements and temperature ranges.
   (b) Why cold junction compensation is required? Explain an automatic cold junction compensation method. [8+8]
4. (a) What way strain gauge be used to measure pressure, explain.
   (b) Explain the construction and working of hydraulic load cell. [6+10]
5. Explain in detail with neat sketches
   (a) liquid level measurement using capacitive transducer.
   (b) Cryogenic fuel level indicator. [8+8]
6. (a) Explain the general theory of seismic instruments used for vibration/accelerator measurement.
   (b) Explain the working of a bonded strain gauge accelerometer with neat sketch. [8+8]
7. (a) What is the need for rosette?
   (b) How dew point temperature is measurement.
   (c) Describe the tension measurement using strain gauge with neat diagram. [4+4+8]
8. Write short notes on the following:
   (a) Servomechanism
   (b) Temperature control system
   (c) Position control system [16]

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Code No: RR220305 **Set No. 3**

II B.Tech II Semester Supplimentary Examinations, Apr/May 2007
INSTRUMENTATION AND CONTROL SYSTEMS
(Common to Mechanical Engineering, Production Engineering and Automobile Engineering)

Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

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1. (a) Discuss briefly on the reliability of measurement systems. Explain how reliability improvement can be achieved.
   (b) "A pricise instrument need not be accurate" explain. [8+8]
2. (a) Explain the construction and working of an A.C. Tachometer generator. Describe its limitations.
   (b) Explain the construction, working of a photoelectric tachometer. Explain its advantages and disadvantages. [16]
3. (a) Explain the principle behind the temperature measurement by radiation methods.
   (b) Explain the working of a total radiation in pyrometer. [16]
4. (a) Describe the construction and working or c-type, spiral type and helical type Bourdon gauges with neat diagrams.
   (b) Describe the construction, working and theory of a diaphragm type strain gauge transducer using four strain gauges. Describe how they are connected in a wheetstone bridge and what is the output obtained. List their advantages and disadvantages. [16]
5. (a) Write the principle of operation of a bubbler purge and float leased level indicator.
   (b) With neat sketch explain any one type of ultrasonic fluid level indicator. [16]
6. (a) Explain the calibration procedure for an accelerometer.
   (b) What are the different methods of converting vibration into a voltage? Explain any on in detail. [8+8]
7. (a) Write the desirable characteristics of bonding material.
   (b) Explain how the strain gauge functions as transducer as deflection and measurement of elastic deformation. [4+12]
8. (a) Draw the schematic block diagram for an automatic washing machine and identify the input and output of the system.
   (b) Draw the block diagram of a biological control system when a human hand approaches to an object to grip it. Discuss the functions of various elements of the system. [11+5]

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Code No: RR220305 **Set No. 4**

II B.Tech II Semester Supplimentary Examinations, Apr/May 2007
INSTRUMENTATION AND CONTROL SYSTEMS
(Common to Mechanical Engineering, Production Engineering and Automobile Engineering)

Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

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1. (a) Define the following:
   i. Measurand
   ii. Primary measuring element
   iii. Measurement
   iv. Calibration
   (b) What do you mean by functional elements? Explain the division of a measurement system into fractional elements with examples. [8+8]
2. What are mechanical Tachometers? Explain with examples. Describe the disadvantages of mechanical Tachometers. [16]
3. (a) Explain the principle behind the bimetallic thermometer with necessary diagram.
   (b) Mention some important application of bimetallic sensors.
   (c) Give various configuration of bimetallic sensors. [16]
4. (a) Explain, how an elastic diaphragm gauge is used to measure pressure with the help of relevant sketch.
   (b) List the limitations of elastic diaphragm gauge.
   (c) Discuss the merits and demerits of elastic sensing elements. [8+4+4]
5. Explain in detail with neat sketches
   (a) liquid level measurement using capacitive transducer.
   (b) Cryogenic fuel level indicator. [8+8]
6. (a) What are the advantages of piezoelectric type accelerometer?
   (b) Why vibrations has to be measured. How vibrations are measured?
   (c) What is a psychrometer? Explain the working of a psychrometer with neat sketch. [4+2+8]
7. (a) What are load cells? Name the application of load cells.
   (b) Describe the working principle of strain gauge bridge with neat sketch. Indicate their arrangements for measurement of torque on a circular shaft. [4+12]
8. (a) What are the basic elements of a control system? Explain.

(b) Describe a control system to fill a tank with water after it is emptied through an output at the bottom. This system automatically stops the inflow of water when the tank is filled. Draw the block diagram of the system. [6+10]

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